

CLIENT DELIVERY DOCUMENT

Hierarchia Manager Portal User Guide

Version: 1.0

Date: May 12, 2026

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Classification: Client Confidential

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1. Purpose of This Document

This document provides a detailed comparison between **Anthropic Claude Sonnet 4.5** and **OpenAI GPT-4o mini** for use inside the **Hierarchia Manager Portal**.

The goal is to help the client understand the practical difference between these two models in terms of:

- Quality
- Efficiency
- Speed
- Cost impact
- Credit usage per message
- Business impact
- Best-fit use cases
- Operational trade-offs inside this project

This comparison is written specifically for the portal's current architecture, where AI is used for:

- Smart AI chat
- Document understanding
- Tool-connected assistant workflows
- AI validation support
- Summaries and operational analysis

2. Project Context

Within the current project architecture:

- **Smart AI Chat default model** is `anthropic/claude-sonnet-4-5`
- **Validation pipeline default model** is currently Gemini-based
- **Fallback validation model** is `openai/gpt-4o-mini`
- **AI credit accounting in the current app is model-agnostic**

That means the current application does **not** charge different internal credits depending on whether a user message uses Claude Sonnet 4.5 or GPT-4o mini.

In the current implementation:

- **1 Smart AI query = 1 credit**
- **1 submission validation = 1 credit**

This is important because the **financial cost to the system is not equal**, even though the internal user credit deduction is currently equal.

3. Executive Summary

If the comparison is reduced to one practical business conclusion, it is this:

- **Claude Sonnet 4.5** should be the **recommended model for this project** because it delivers stronger reasoning, higher-confidence answers, and better first-pass usefulness for the type of business questions the portal is designed to support
- **GPT-4o mini** remains the better model only when the priority is **minimum cost per message, very high message volume, or strict budget control**

In simple terms:

- **Claude Sonnet 4.5** gives better answer quality, better first-response quality, and better practical efficiency for management and decision-support usage
- **GPT-4o mini** gives better raw cost efficiency and faster low-cost response generation

So the real decision is not only “which model is cheaper?”

The real decision is:

- Do you want the model that produces the **strongest and most useful answer on the first attempt?**
 - Or do you want the model with the **lowest raw API cost per message?**
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4. Side-by-Side Comparison Table

Category	Claude Sonnet 4.5	GPT-4o mini
Overall quality	Excellent	Good to very good
Deep reasoning	Excellent	Moderate
Tool use / assistant behavior	Excellent	Very good
Document analysis	Excellent	Good
Complex multi-step prompts	Strong	Acceptable but less reliable
Response speed	Fast	Very fast
Raw API cost per message	High	Very low
Raw cost efficiency	Low	Excellent
First-pass answer efficiency	Excellent	Good
Business decision efficiency	Excellent	Moderate
High-volume deployment suitability	Good	Excellent
Best role in this project	Recommended primary Smart AI model	Secondary budget or fallback model

5. Quality Comparison

5.1 Reasoning Quality

Claude Sonnet 4.5

Claude Sonnet 4.5 is better suited for:

- Complex reasoning
- Multi-step interpretation
- Policy-heavy answers
- Comparative analysis
- Higher-confidence synthesis across multiple facts
- More polished final responses

It is especially strong when the user asks questions such as:

- Compare multiple teams and explain performance differences
- Analyze submission quality patterns and identify causes
- Interpret business context across tasks, validation summaries, and activity history
- Produce clearer decision-support style responses

GPT-4o mini

GPT-4o mini performs well for:

- Everyday assistant usage
- Short and medium complexity prompts
- Basic summaries
- Routine Q&A
- Lightweight operational questions

It is weaker than Sonnet when prompts require:

- Long-chain reasoning
- High nuance
- Careful judgment across several data points
- Strongly structured business interpretation

Practical conclusion

- If the project needs **better intelligence quality**, Claude Sonnet 4.5 is stronger
 - If the project needs **acceptable daily answers at very low cost**, GPT-4o mini is stronger
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5.2 Document Understanding Quality

For this portal, document understanding matters because the AI layer often works with:

- Submission summaries
- Validation context
- Retrieval content from indexed records
- Operational records and portal data

Claude Sonnet 4.5

Strengths:

- Better at reading subtle meaning from longer text

- Better at identifying missing logic or weak evidence
- Better at producing higher-quality explanations
- Better at maintaining context across longer conversations

GPT-4o mini

Strengths:

- Good enough for short and moderate document queries
- Good at fast summarization for standard cases
- Effective for simple operational retrieval questions

Limitations:

- More likely to simplify complex context too aggressively
- Less strong than Sonnet in nuanced reasoning
- Less ideal when the answer quality directly affects stakeholder confidence

Practical conclusion

If the Smart AI assistant is positioned as a **high-value advisory tool**, Sonnet is a better fit.

If it is positioned as a **fast internal productivity helper**, GPT-4o mini is often enough.

5.3 Tool Calling and Assistant Workflow Quality

In this project, the Smart AI assistant is not just a plain chatbot. It can work with:

- Portal data
- Retrieval context
- Tool-connected assistant actions
- Threaded conversations
- Usage tracking

Claude Sonnet 4.5

Claude Sonnet 4.5 is better when the assistant needs to:

- Decide what information to use
- Ask for the right tool outputs
- Chain multiple steps more carefully
- Produce a more coherent final answer after tool usage

GPT-4o mini

GPT-4o mini is still strong for tool-connected workflows, especially when:

- The question is simple
- The tool outputs are already clear
- The answer does not require heavy synthesis

Practical conclusion

- For **higher-trust assistant behavior**, Sonnet is better
 - For **cheap and fast tool-assisted responses**, GPT-4o mini is better
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6. Efficiency Comparison

6.1 Response Speed

Approximate project-aligned estimates place both models in the “fast” range, but GPT-4o mini is usually faster and cheaper for general response generation.

Measure	Claude Sonnet 4.5	GPT-4o mini
Time to first token	~600–1200 ms	~300–500 ms
Relative speed	Fast	Very fast
Best for live-heavy usage	Good	Excellent

Practical conclusion

GPT-4o mini is more efficient when the client expects:

- Higher message volume
- Faster visible replies
- Lower waiting time in routine interaction

Claude Sonnet 4.5 is still fast, and for this project raw latency alone should not define efficiency. In higher-value workflows, Claude often becomes more efficient in practice because it is more likely to produce a stronger answer in the first response, reducing follow-up prompting and clarification cycles.

6.2 Raw Cost Efficiency

Using the project's existing AI strategy assumptions:

- Approximate **input per Smart AI message**: 2,000 tokens
- Approximate **output per Smart AI message**: 500 tokens

Approximate model cost per message:

Model	Approximate Cost per Message
GPT-4o mini	~\$0.0006
Claude Sonnet 4.5	~\$0.0135

Approximate monthly cost at 1,000 messages:

Model	Approximate Monthly Cost
GPT-4o mini	~\$0.60
Claude Sonnet 4.5	~\$13.50

Cost ratio

Using the above assumptions:

- **Claude Sonnet 4.5 is approximately 22.5× more expensive per message than GPT-4o mini**

Practical conclusion

This is the biggest raw commercial difference between the two models.

GPT-4o mini is dramatically more efficient from a pure token-cost perspective.

6.3 Practical Work Efficiency

For this project, practical efficiency should also be measured by how effectively the model completes higher-value work with fewer extra turns.

On that basis, Claude Sonnet 4.5 is stronger because it more often provides:

- Better first-pass answers
- Better synthesis across multiple business facts
- Better decision-support quality for managers and stakeholders
- Better handling of nuanced or multi-step questions

- Lower risk of needing repeat prompts to get a satisfactory answer

In a management-facing system, this matters because one strong answer can be operationally more efficient than multiple cheaper but weaker answers.

That means:

- **GPT-4o mini** is more efficient for **raw scale**
- **Claude Sonnet 4.5** is more efficient for **high-value output quality and first-pass usefulness**

Since this portal is positioned around AI-assisted decision support, document understanding, and management workflows, **Claude Sonnet 4.5 is the more practically efficient and recommended model for this project.**

7. Credit Usage per Message

7.1 Current Credit Logic in This Project

The current application logic is simple:

- **1 Smart AI query = 1 credit**
- **1 submission validation = 1 credit**

This is true regardless of the actual underlying model.

So in the current system:

Action	Claude Sonnet 4.5	GPT-4o mini
1 Smart AI message	1 credit	1 credit
1 validation event	1 credit	1 credit

Important implication

From the user's point of view, both models cost the same in internal credits.

But from the organization's point of view, they do **not** cost the same financially.

7.2 Business Interpretation of Current Credit Logic

The current credit system is good for:

- Simple user understanding

- Easy quota administration
- Clean reporting
- Consistent internal usage tracking

But it does **not** reflect actual model cost differences.

This means that under the current policy:

- A user can spend **1 credit** on a cheap GPT-4o mini response
- Another user can spend **1 credit** on a much more expensive Claude Sonnet 4.5 response

So the same internal credit can represent very different real monetary value.

7.3 Recommended Weighted Credit Interpretation

If the client wants a credit system that better reflects actual model cost, a weighted approach can be introduced in the future.

Simple recommended baseline

Use GPT-4o mini as the cost baseline:

- **GPT-4o mini = 1 credit unit**

Then translate Claude Sonnet 4.5 proportionally:

- **Claude Sonnet 4.5 $\approx$ 22 to 25 credit units** under the same token assumptions

Suggested practical rounded model

A cleaner business-friendly version could be:

Model	Suggested Weighted Credit Value per Message
GPT-4o mini	1
Claude Sonnet 4.5	20

This is **not currently implemented** in the application.

It is a commercial and governance recommendation only.

7.4 Recommendation on Credit Policy

If simplicity is the priority:

- Keep the current policy of **1 message = 1 credit**

If cost control is the priority:

- Introduce model-weighted credits

If project quality and client-facing AI experience are the priority:

- Keep **Claude Sonnet 4.5** as the recommended default Smart AI model
- Use the current simple credit system unless differentiated commercial plans are introduced later

If premium AI access segmentation is required later:

- Keep Claude Sonnet 4.5 for core recommended usage
 - Use GPT-4o mini as a secondary low-cost option for budget-sensitive or non-critical usage bands
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8. Impact Comparison

8.1 User Experience Impact

Claude Sonnet 4.5

User-facing impact:

- Better answer depth
- Better professional tone
- Better explanation quality
- Stronger trust in AI output
- Better performance on harder questions
- Better chance of resolving a query well on the first attempt

Risks:

- Higher running cost
- Lower budget scalability
- More expensive for large user populations

GPT-4o mini

User-facing impact:

- Fast responses
- Good enough for many daily tasks

- Lower-cost availability to more users
- Easier large-scale rollout

Risks:

- Lower answer depth for complex questions
 - More limited reasoning quality on high-value analytical prompts
 - May feel less premium in expert or executive use cases
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8.2 Operational Impact

Claude Sonnet 4.5

Operationally better for:

- Recommended primary AI assistant experiences
- Higher-value knowledge work
- Clients who care more about answer quality, trust, and decision support than raw message volume
- Lower-volume but more important AI interactions
- Situations where stronger first-pass output reduces operational rework

GPT-4o mini

Operationally better for:

- Broad access across many users
 - Routine support and daily productivity
 - High-frequency usage patterns
 - Tight AI budgets
 - Controlled cost growth
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8.3 Financial Impact

This is where the difference is most dramatic.

Claude Sonnet 4.5

Financial profile:

- Premium quality

- Premium cost
- Best used where answer quality directly matters

GPT-4o mini

Financial profile:

- Much lower per-message cost
- Better cost predictability at scale
- Safer for wider rollout and high concurrency

Practical conclusion

If the client wants AI everywhere for everyone at the lowest cost, GPT-4o mini is financially easier to justify.

If the client wants AI as a high-confidence business tool, Claude Sonnet 4.5 is easier to justify despite the higher raw cost.

9. Best Use Cases in This Project

9.1 Best Use Cases for Claude Sonnet 4.5

Claude Sonnet 4.5 is a better fit when the user asks:

- Complex comparative questions
- Team performance interpretation
- Multi-factor operational reasoning
- Long-form assistant analysis
- High-trust business questions
- Questions where answer quality matters more than message cost

Examples:

- Compare three departments and explain why one has a lower pass rate
 - Review submission trends and suggest management actions
 - Analyze repeated failure patterns and summarize the likely root causes
 - Produce a more executive-level answer from several sources of context
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10. Strengths and Weaknesses Summary

Claude Sonnet 4.5

Strengths

- Excellent reasoning
- Better final answer quality
- Strong tool-use orchestration
- Better at subtle document interpretation
- Better for recommended Smart AI experiences
- Better first-pass usefulness for complex business questions

Weaknesses

- High cost per message
- Lower raw cost efficiency
- Harder to scale broadly without stronger quota control

10.2 GPT-4o mini

Strengths

- Extremely cost-efficient
- Very fast
- Excellent for scaled usage
- Good baseline assistant model
- Strong choice for fallback and budget deployments

Weaknesses

- Lower reasoning depth than Sonnet
 - Lower premium quality on difficult prompts
 - More likely to feel “good enough” rather than “best-in-class”
 - More likely to need follow-up prompts on higher-complexity questions
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11. Recommendation Options

Option A — Recommended Project Mode

Use **Claude Sonnet 4.5** as the Smart AI model when:

- The assistant is expected to deliver stronger business value
- The client wants higher-confidence Smart AI output
- The AI is expected to handle harder analytical work
- The portal is being positioned as a premium, decision-support capable system

This is the **recommended mode for this project**.

Option B — Cost-Optimized Mode

Use **GPT-4o mini** as the Smart AI model when:

- The client wants broad rollout
- The team is large
- Budget matters strongly
- Most AI prompts are routine rather than complex

Option C — Hybrid Policy

This is the best secondary option if the client later wants tighter budget segmentation.

Suggested hybrid approach:

- Keep **Claude Sonnet 4.5** as the default recommended Smart AI model
- Use **GPT-4o mini** only for:
 - budget-sensitive usage bands
 - very high-volume low-complexity usage
 - fallback or secondary cost-control cases

This gives the client:

- premium quality as the main experience
 - optional cost control where needed
 - better control over AI budget without lowering the default Smart AI quality
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12. Final Decision Matrix

Priority	Better Choice	Reason
Highest answer quality	Claude Sonnet 4.5	Better reasoning and better synthesis
Lowest cost per message	GPT-4o mini	Much cheaper
Best first-pass answer quality	Claude Sonnet 4.5	Better chance of solving the query well in one response
Recommended model for this project	Claude Sonnet 4.5	Best fit for management, AI-assisted decision support, and premium Smart AI value
Fastest low-cost rollout	GPT-4o mini	Best raw scale efficiency
Premium Smart AI experience	Claude Sonnet 4.5	Higher-value responses
Broad low-budget organization rollout	GPT-4o mini	Budget-safe at volume
Executive or difficult analysis	Claude Sonnet 4.5	Better nuance and judgment
Fallback model	GPT-4o mini	Cheap, fast, dependable
Hybrid enterprise model strategy	Both	Keep Claude as primary, GPT-4o mini as secondary budget layer

13. Final Conclusion

For the **Hierarchia Manager Portal**:

- **Claude Sonnet 4.5** is the stronger model for **quality, reasoning, business interpretation, first-pass usefulness, and overall project fit**
- **GPT-4o mini** is the stronger model for **raw cost efficiency, speed, and low-cost scale**

If the client asks which model is better for this project, the answer is:

- **Claude Sonnet 4.5**

If the client asks which model is cheaper per message, the answer is:

- **GPT-4o mini**

If the client asks what should be recommended as the primary Smart AI model, the answer is:

- **Claude Sonnet 4.5**

If the client later wants a cost-control layer, then the secondary strategy can be:

- Keep Claude Sonnet 4.5 as the default
 - Use GPT-4o mini only where lower-cost secondary usage is needed
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14. Important Note on Pricing and Rates

All per-message cost estimates in this document are **approximate** and based on the existing strategy assumptions already documented for this project.

Actual cost can vary based on:

- Provider pricing updates
- OpenRouter rate changes
- Input token length
- Output token length
- Tool usage patterns
- Retrieval context size
- Conversation history length

For production budgeting, the client should always treat these numbers as planning estimates and verify the latest live rates before final commercial decisions.